Exam Date & Time: 30-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V SEM B. Tech (BME) DEGREE END SEMESTER MAKE-UP EXAMINATIONS, DEC-JAN -2024-25

MEDICAL DEVICES [BME 3122]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) Illustrate a breathing therapy unit that uses intermittent breaths to inflate the lungs. (4)

1B) Explain the working principle of manual artificial ventilators and indicate their drawbacks compared to its automated counterparts. (3)

1C) Explain, with a neat figure, the working principle of flame photometry. (3)

2A) What is hyperbilirubinemia? What are the problems caused by hyperbilirubimia? Determine how the levels of bilirubin can be monitored in neonates non-invasively. (5)

2B) List the advantages of thermography over other methods and also, compare the different detectors used in thermographs. (3)

2C) The speed of sound in the human body is 1500m/s. If a fetus is 6cms below the mother's skin, how long will it take for the echo to be received? (2)

3A) Why is Time-Gain Compensation (TGC) required in ultrasound? With a suitable TGC curve, discuss the various TGC controls that are normally adjustable by the sonographer. (4)

3B) Illustrate how electrical conductivity change can be used for blood cell counting. (4)

3C) In a blood cell coulter counter, the threshold is set to zero and the output display reads 5.31×10^{12}/ liter. The threshold is then set to T1 and the output reading becomes 5.18×10^{12}/ liter. The threshold is then set to T2 and the display shows 0.18×10^{12}/ liter. Find the count of RBC, WBC and platelets in units of cells/ liter. [Assume: $0 < T1 < T2$]. (2)

4A) Illustrate a type of oxygenator where bubbles are used to increase the gas transfer efficiency. Also, mention the drawbacks of using this type of oxygenators. (4)

4B)	Explain the characteristics of the ideal heart-lung machine.	(4)
4C)	Compare B-scan and M-scan of ultrasound imaging.	(2)
5A)	Illustrate with a neat figure, the basic functions performed by the dialysis machine.	(4)
5B)	Explain the different steps involved in the life cycle of a medical device.	(3)
5C)	Describe the three classifications of medical devices, providing examples for each category.	(3)

-----End-----